ModelGate

Intelligent AI Routing — Built from Your Contracts

One line changed. Millions of premium calls rerouted.

FastAPI + Python · Next.js + TypeScript · Docker

Agents Assemble:

Aaryan Kapoor - Back End
Pradyumna Kumar - Front End
Danny Tran - Slides

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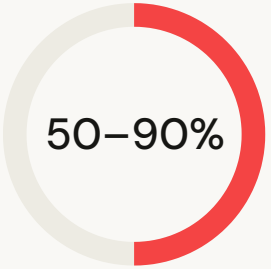
The Assurant Way

Helping people thrive in a connected world.

The connected world runs on AI. AI is expensive, wasteful, and impossible to configure correctly at scale.

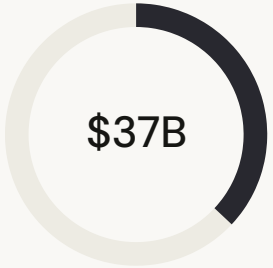
Energy Difference

More energy consumed by premium models vs small models per query



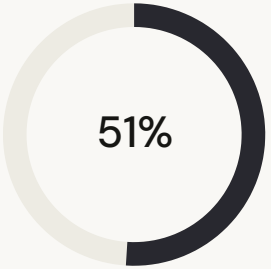
Wasted Spend

Of enterprise AI inference spend wasted on over-provisioned models



Enterprise Spend

Enterprise GenAI spend in 2025 up 3.2× year-over-year



ROI Measurement

Of organizations can actually measure their AI ROI

> "Without security and efficient application infrastructure, we are at risk of increasing the waste and consumption with our day-to-day tools." — Assurant Challenge

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The Problem Is Universal

Every layer of the AI stack is affected.

Affected Party	The Problem Today	Consequence
Developers & Product Teams	No time to evaluate 200+ models. Pick one **premium option** and never revisit it.	Overpay by 10–30×
Enterprise Customers	Contract constraints manually translated into config — or **silently ignored**. Compliance risk.	Legal exposure
End Users	"What is your return policy?" routed to a **slow premium model**. Waits 2 seconds for a simple answer.	Poor experience
AI Data Centers	**Unnecessary load** on premium GPU clusters from queries that a small model would answer perfectly.	Wasted capacity
The Environment	Premium models: **~39 Wh/query** Small models: **~0.22 Wh/query** . A **180×** energy difference per call.	415 TWh/yr and rising

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The Fix

One line of code. We handle everything else.

Before — Every Team Today

```
# Direct provider call
from openai import OpenAI
client = OpenAI(
    api_key="sk-...",
    base_url="https://api.openai.com/v1"
)
# Same model. Every request. Forever.
```

After — With ModelGate

```
# Same code. Different URL.
from openai import OpenAI
client = OpenAI(
    api_key="",
    base_url="https://assurant.com/samsung/v1"
)
# Right model. Every request. Auto.
```

API Compatible

Full OpenAI spec — no SDK changes, no rewrites.

Policy Enforced

Contract constraints become routing rules, automatically.

Always Optimal

Cheapest model that satisfies every constraint, per request.

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Who It Serves

Three Personas. One Solution. Simultaneous benefit.

Developer / Product Team

"I want the best AI model. I don't have the time to evaluate every new model"

Enterprise Compliance

"Our contracts ban certain providers. I'm not sure engineers will remember all the time"

Operations / Platform

"I have no idea what our AI is costing us or which models we're using"

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System Architecture

Two phases. Fully automated. No ongoing effort.

Phase 1: Onboarding

≈ 30 seconds

- Upload contract (PDF/text/SLA docs)
- LLM reads and extracts constraints
- Structured Customer AI Profile generated
- OpenAI-compatible endpoint goes live

Phase 2: Runtime Routing

~50ms overhead

- Prompt received and classified
- Heuristic filters applied
- Models scored and ranked
- Best model selected and response routed

Simple

Ex. Models: GPT 5.4 nano · Gemini 3.1 Flash Lite · Haiku 4.5

Use: FAQ, greetings, status checks

Medium

Ex. Models: GPT 5.4 mini · Sonnet 4.6 · Grok 4.1 Fast

Use: Summarize, explain, troubleshoot

Complex

Ex. Models: GPT 5.4 · Opus 4.6 · Gemini 3.1 Pro · Grok 4.20

Use: Multi-doc reasoning, code gen, legal

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Live Demo 🚀

Five steps. One contract. Immediate routing.

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Evidence

ModelGate Live Results

<30s	~50ms	%	%
Onboarding	**Classification**	**Simple %**	**Savings vs Always-Premium**
Contract → endpoint	Arch-Router-1.5B	Fill after demo	Fill after demo

"Smart routing delivers 85–98% cost reduction while maintaining quality indistinguishable from always-premium."

Up to 98% cost reduction while matching GPT-4 quality by cascading through cheaper models. *FrugalGPT (Stanford, 2023)*

>85% cost reduction at 95% of GPT-4 performance using a trained routing model. *RouteLLM (LMSYS, ICLR 2025)*

Global data center consumption: 415 TWh in 2024. Projected to exceed 1,000 TWh by 2026. *IEA Energy Report (2025)*

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Evidence

Published Research

Source	Finding	Validation
FrugalGPT (Stanford, 2023)	Up to 98% cost reduction while matching GPT-4 quality by cascading through cheaper models.	Academic peer-reviewed
RouteLLM (LMSYS, ICLR 2025)	>85% cost reduction at 95% of GPT-4 performance using a trained routing model.	Conference published
RouteLLM Matrix Factorization	Only 14% of queries actually require the most capable model. 86% can use smaller ones.	Data-driven analysis
IEA Energy Report (2025)	Global data center consumption: 415 TWh in 2024. Projected to exceed 1,000 TWh by 2026.	Industry standard

ModelGate Live Results

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Arch-Router-1.5B

%
Simple %
Fill after demo

%
Savings vs Always-Premium
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Model Fine-tuned

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Impact & Close

One change. Benefits at every layer of the stack.

Business

60–98% cost savings on AI inference through intelligent right-sizing

60–98% saved

End Users

Simple queries get fast answers. Complex queries get powerful models.

Right model, always

Data Centers

Less unnecessary GPU load on premium clusters. Resources freed.

Less waste

Energy Grid

Up to 180× less energy per routed-away premium call. Compounding effect.

180× difference

Compliance

Contract constraints become routing rules. Enforced per request, automatically.

Zero human error

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> "ModelGate is the lowest-friction AI optimization available today. You change one line of code. We eliminate thousands of wasteful premium model calls automatically."

Thank you.

KSU · Social Good Hackathon · 2026

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